

IA-840F

OFS 2026.1 and AHLS Support Guide

From a new Linux installation to
building and running SYCL vector-add

FPGA platform	BittWare IA-840F, Agilex 7, AGFB027R25A2E2V
Static platform	OFS ofs-2026.1-1 FIM and matching PIM/PR collateral
FPGA implementation	Quartus Prime Pro 26.1.1 Build 130 (26.1 family)
Compiler/runtime	Altera HLS IP Gen (AHLS) 2026.1.0
Host access stack	Latest published OFS OPAE release 2.14.0-3
Board package	Current IA-840F oneAPI Accelerator Support Package
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Scope

A practical, version-pinned guide for the current project stack. It covers one-time host and board preparation, the normal compiler-driven BSP link, the four standard/USM and plain/kclk configurations, and a verified prebuilt-image path.

This is project documentation, not an official BittWare or Altera publication. The supplied BittWare support guide informed the chapter structure; its older installation and programming recipes are not substituted for this stack.[\[1\]](#)

Contents

Read this before starting	iv
1 Overview and included configurations	1
1.1 What the stack does	1
1.2 Variants and compile flows	1
1.3 Version and identity contract	1
2 Inputs, workspace and host prerequisites	3
2.1 Required files	3
2.2 Use explicit paths	3
2.3 Host, power and cooling	3
2.4 BIOS, IOMMU and huge pages	4
2.5 Build resources and limits	4
3 Installing the current software stack	5
3.1 System build dependencies	5
3.2 Quartus Prime Pro 26.1.1	5
3.3 AHLS 2026.1.0	5
3.4 BittWare SDK and IA-840F product support	6
3.5 OPAE 2.14.0-3: build into an owned prefix	6
3.6 Linux DFL driver on Rocky 9.8	7
4 The OFS 2026.1 FIM and matching PIM platform	9
4.1 Use the paired IA-840F platform	9
4.2 Already-provisioned versus factory-programmed card	9
4.3 Permanent provisioning: conditional operator procedure	10
4.4 Activation and postboot verification	10
4.5 Rebuilding the FIM/PIM is a maintainer operation	11
5 Installing and selecting the IA-840F ASP	12
5.1 Extract and verify the release	12
5.2 Directory map	12
5.3 Compiler environment: source AHLS, then select Quartus	12
5.4 Private runtime activation	13

6	Preparing the application VF and card access	14
6.1	Discover, do not copy, hardware identifiers	14
6.2	Per-boot VF creation and explicit VFIO selection	14
6.3	Memory lock, huge pages and ownership	14
7	Building vector-add through the BSP	16
7.1	Select one coherent build configuration	16
7.2	Normal full hardware compilation	16
7.3	The native compile/link pattern	16
7.4	Verified prebuilt-image link: fastest initial bring-up	17
7.5	Find the result and inspect it	17
8	Loading and running the FPGA program	18
8.1	Pre-run checklist	18
8.2	Preserve the selected runtime under privilege	18
8.3	CMake run target	19
8.4	What success means	19
9	Validating all four configurations	20
9.1	Use distinct products and one owner at a time	20
9.2	Qualification record for this release	20
9.3	Optional short numerical probes	20
10	Debugging and known operating limits	22
10.1	Wrong Quartus generation or missing FPGA header	22
10.2	Python development mismatch during OPAE configuration	22
10.3	Cannot open the device or user-clock UIO node	22
10.4	No VF after reboot	22
10.5	Timing failure, ROM failure or missing base outputs	22
10.6	Native reuse declines the image	22
10.7	The retained VF lifecycle warning	23
10.8	Failure and recovery discipline	23
11	Maintenance, package integrity and source boundaries	24
11.1	Keep a coherent runtime	24
11.2	Rebuilding MMD/MPF	24
11.3	Image naming and archive validation	24

11.4 What is outside the validated matrix	24
12 Operator checklist	25
Sources and qualification evidence	26

Read this before starting

One-time setup and routine application work are different operations. A new host needs the Linux access stack, toolchain, board power/cooling and a matching static FIM. Once those are correct, vector-add compilation and an AFU partial-reconfiguration load do not require another full-device flash or a workstation reboot.

Two routes are documented. The full hardware build compiles your SYCL kernel into a new image. The prebuilt route links the supplied, already-fitted program through the actual AHLS driver, then runs that image. Use the latter for an initial full-oracle numerical bring-up without spending another FPGA fit; use the former when changing device code. Both produce a real SYCL `.fpga` fat binary, not a raw-AFU replacement host.[\[11\]](#)

The main instructions target one coherent current stack. OPAE is pinned to the latest published release, 2.14.0-3, commit `e4b329185d2c462f5b8c3c74443a5c5cdb25f3b3`. This is a deliberate release pin, not an instruction to track a moving development branch.[\[17\]](#)

Commands are recipes, not permission to run a dangerous operation. Perform remote workstation work in an owned `tmux` session. Resolve the actual card, BDFs, IOMMU groups and current ownership before an operation that opens the accelerator, programs an image, creates a VF or changes power. An unknown process or timed-out hardware access is a stop condition, not a reason for another reset, PR load or probe.[\[10\]](#)

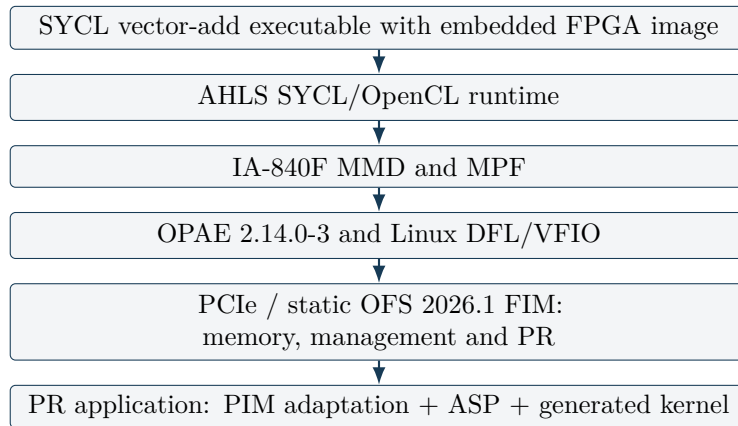
Obtain the complete input set. The ASP archive includes the application images and compiler PR template. It is not itself a factory-card cold-boot flash package. A factory-programmed card additionally needs the matched, verified full-device FIM provisioning bundle described in Section 4.[\[8\]](#)

Copying commands. Define the path variables in Section 2 first. Replace example hardware identifiers only with identifiers verified for your host. Wrapped listings display a small “continuation” marker; it is a layout aid, not a token to enter in a shell.

1 Overview and included configurations

1.1 What the stack does

AHLS compiles SYCL device code and supplies the `kernel_system`. The oneAPI ASP provides board-facing kernel, memory, CSR, clock/reset and runtime integration. The ASP and generated kernel together form the application AFU inside the static FIM's PR region. PIM adapts the AFU to the FIM interfaces; it is not a separately loaded application tier. On the host, SYCL/OpenCL reaches the accelerator through MMD/MPF, OPAE and the Linux DFL/VFIO drivers.^[3]



Logical software/hardware path, not a physical floorplan or a latency diagram.

1.2 Variants and compile flows

Variant	Flow	Intended use
<code>ofs_ia840f</code>	<code>afu_flat</code>	Standard local-memory/buffer path. Start here.
<code>ofs_ia840f</code>	<code>afu_flat_kclk</code>	Same memory variant with the kernel-clock/PIM-CDC alternative.
<code>ofs_ia840f_usm</code>	<code>afu_flat</code>	USM-enabled variant; the example uses shared USM.
<code>ofs_ia840f_usm</code>	<code>afu_flat_kclk</code>	USM with the alternate clock topology.

Both variants target the same IA-840F part and two 16-GiB local DDR banks (32 GiB aggregate). The USM variant adds the host/shared-memory path. The `_kclk` choice enables `USE_KERNEL_CLK_EVERYWHERE_IN_PR_REGION` in the PR integration and changes its PIM CDC/domain configuration; it does not change the static PCIe or DDR reference clocks. Output revision names alone are not proof of topology.^[20]

1.3 Version and identity contract

The static FME interface UUID is `fc603c44-5c8f-5e94-bcbe-a5780030947c`. The standard and USM AFU UUIDs are `51ED2F4A-FEA2-4261-A595-918500575509` and `5D9FEF7B-C491-4DCE-95FC-F979F6F061BE`, respectively. FME and AFU UUIDs serve different roles. A matching AFU UUID is not enough to identify the physical card, and the two compile flows of one variant share an AFU UUID.^[20]

The package is a project-qualified IA-840F port of current OFS ASP machinery. A stock N6001 board package, an unrelated static QDB, or an older vendor BSP is not an interchangeable dependency.[\[20\]](#)

2 Inputs, workspace and host prerequisites

2.1 Required files

Obtain these inputs before installation. Licensed tools and full-device programming images are separate from a source checkout.

Input	Purpose
Quartus Pro 26.1.1 + Agilex 7 support	FPGA implementation and programming-file tools.
AHLS 2026.1.0 distribution/helper	Compiler, SYCL/OpenCL runtime and FPGA tools.
Valid Quartus entitlement	Required for the implementation operations you use.
BittWare SDK 2026.1 EL9 package	Current IA-840F management/product support.
OPAE 2.14.0-3 sources	Host libraries, plugins, utilities and platform tools.
IA-840F ASP release archive	Current compiler metadata, RTL, runtime and four image sets.
Paired FIM provisioning bundle	Full-device SOF/RPD/MAP and recovery material for a factory card.
Matching PR platform/template	The accepted static QDB and PIM/compiler collateral.

The latest OPAE release is available at <https://github.com/OFS/opae-sdk/releases/tag/2.14.0-3>.^[17] Obtain the proprietary SDK and board-revision hardware documentation from the BittWare developer distribution; do not invent a download URL or reuse a package intended for a different board.

2.2 Use explicit paths

These variables are used throughout the guide. The example layout keeps sources, tools, build trees and logs separate.

```
WORK="$HOME/ia840f-work"
DOWNLOADS="$WORK/downloads"
AHLS="$HOME/ahls/altera_hls/aclsycl"
QUARTUS="/opt/altera/26.1.1/quartus"
BSP="$WORK/asp/ia840f"
OPAE_SRC="$WORK/src/opae-sdk"
OPAE_BUILD="$WORK/build/opae"
OPAE_PREFIX="$WORK/install/opae"
PR="$BSP/pr_build_template"
mkdir -p "$DOWNLOADS" "$WORK/src" "$WORK/build" "$WORK/logs"
tmux new-session -s ia840f-setup
```

Run the remaining workstation commands inside that session. The unpacked BSP must be writable by its operator because private runtime activation creates prefix-specific FCD/ICD files. PR can instead name a separately supplied, matching accepted PR export; do not point it at arbitrary FIM source.

2.3 Host, power and cooling

The qualified host uses Rocky Linux 9.8, x86-64, kernel 5.14.0-687.48.1.el9_8.x86_64; its compatible DFL modules are exposed through enterprise **weak-updates**. This is project validation, not a claim that every distribution is vendor-certified.^[5]

Install the card, auxiliary power and cooling according to the hardware guide for the physical board

revision. Power the host down before physical installation. The PCIe slot alone is not a substitute for the required auxiliary supply. USB IN is useful for BMC management/provisioning; it is not required for routine PCIe PR merely because a historical guide used JTAG. The qualification workstation negotiates Gen3 x16; do not attempt unrelated link retraining to force a Gen4 result.

2.4 BIOS, IOMMU and huge pages

Enable the host's IOMMU/virtualization and SR-IOV features. Use a deliberate Secure Boot/module-signing policy: unsigned out-of-tree modules require a compatible policy, not an automatic security bypass. The verified host reserves 2048 two-MiB huge pages and enables Intel IOMMU and PCI resource reallocation.^[5] For the Intel/Rocky qualification configuration:

```
sudo grubby --update-kernel=ALL \
  --args='intel_iommu=on pci=realloc hugepagesz=2M hugepages=2048'
# Reboot only after completing the intended host setup and saving work.
# After returning to an owned tmux session:
cat /proc/cmdline
grep -E '^HugePages_|^Hugepagesize:' /proc/meminfo
```

Use the appropriate IOMMU option for another CPU platform; do not copy `intel_iommu` onto a different architecture without checking its Linux support. The huge-page reservation is a setup choice, not proof that allocation and access work; the numerical run is the verification step.

2.5 Build resources and limits

CMake 3.20 or later is required by the BSP/sample targets. The observed qualification host has approximately 125 GiB usable RAM. That observation is not an Altera minimum specification. Use live headroom to admit builds and reserve a 55-GiB per-process virtual-address-space limit:

```
ulimit -v 57671680
free -h
df -h "$WORK"
```

At most two fits run simultaneously in this workflow. A per-process virtual limit does not constrain aggregate RSS or all children as one budget. A prebuilt image link is much smaller than a full fit.

3 Installing the current software stack

3.1 System build dependencies

On a new Rocky host, install the build dependencies through the configured Rocky repositories. Review the transaction before accepting it:

```
sudo dnf install \
  gcc gcc-c++ make cmake git tmux tar gzip unzip \
  python3 python3-pip python3-devel \
  libuuid-devel json-c-devel libedit-devel \
  numactl-devel hwloc-devel tbb-devel \
  openssl-devel libffi-devel pkgconf-pkg-config
```

DFL source builds additionally require headers/development files matching the running kernel. AHLS helper prerequisites include `rpm2cpio` and `uidmap`; use the appropriate distribution packages/provider for those commands.^[2] Do not import an unrelated repository key or disable signature checks to work around a failing optional repository. Scope an individual package operation to the already-configured official repositories when necessary.

3.2 Quartus Prime Pro 26.1.1

Install Quartus Pro 26.1.1 Build 130 and Agilex 7 device support in a distinct prefix. Obtain the licensed installer from Altera, inspect its actual help, and select the correct device family. Existing versions may remain installed; this guide selects the build tool explicitly rather than changing the login shell's global default.^[20]

For Altera's `qinst`/Makeself installer, pass `qinst` arguments after the wrapper's `--` separator. A representative component selection is:

```
QINST="/absolute/path/to/the/downloaded/qinst-installer.run"
"$QINST" --help
sudo "$QINST" --noprogess --nox11 -- \
  --cli --accept-eula \
  --download-dir "$DOWNLOADS/quartus" \
  --install-dir /opt/altera/26.1.1 \
  --components quartus,agilex7
# Verify the actual installed version, without a stale override:
env -u QUARTUS_ROOTDIR_OVERRIDE \
  "$QUARTUS/bin/quartus_sh" --version
```

The precise downloaded installer filename and entitlement are supplied by your software distribution. Questa is optional for the hardware vector-add route; install the matched FPGA Edition if you separately require simulation. Set `LM_LICENSE_FILE` to your valid entitlement/server without placing license contents or credentials in this guide.

3.3 AHLS 2026.1.0

The Altera installation helper can install or reuse Quartus and construct an Apptainer-based development environment. Its interactive prompts accept the install directory, Quartus tree/archive and an AHLS package/existing tree containing `ac1sycl/fpgavars.sh`.^[2]

```
cd "$DOWNLOADS"
bash ./ahls-install-helper.sh --help
bash ./ahls-install-helper.sh
# Choose AHLS 2026.1.0, not an unpinned latest compiler.
# For an existing Quartus installation, select /opt/altera/26.1.1.
# Confirm the summary before accepting installation.
```

The helper normally creates `<install>/bin/ahls-sh`. That shell is useful for its documented container flow, but the validated ASP application workflow runs the installed AHLS tools natively on the host after sourcing `fpgavars.sh`. Do not inherit a helper's stale Quartus bind or expect host OPAE/Python installations to appear automatically inside a container.[\[2\]](#)

```
source "$AHLS/fpgavars.sh" >/dev/null
"$AHLS/bin/ahls" --version
test -f "$AHLS/include/sycl/ext/altera/fpga_extensions.hpp"
test -f "$AHLS/host/linux64/lib/libalteracl.so"
```

Require AHLS 2026.1.0 Build 461da9be608f74678d9c52a2dc1cbb67c58d57fa. A version/help result establishes installation visibility, not a successful FPGA board build.[\[20\]](#)

3.4 BittWare SDK and IA-840F product support

Install the current EL9 SDK package from the board distribution. The inspected SDK includes the IA-840F/Agilex product wheel `bw_agilex_product-0.1.49`; product support is not absent merely because an obsolete separate CSP package is no longer present.[\[6\]](#)

```
SDK_RPM="$DOWNLOADS/bittware-sdk-2026.1.0-1.el9.rpm"
rpm --checksig "$SDK_RPM"
sudo dnf install "$SDK_RPM"
sudo usermod -aG bittware "$USER"
# Log out/in when group membership changes.
/usr/share/bittware-sdk/bin/bw_pip install
/usr/share/bittware-sdk/bin/bw_ls
```

Use the executable paths printed/installed by your SDK. Management tools and `bittware-vfiod` are required for the selected management operation, not a reason to start a competing daemon while an application owns the card.

3.5 OPAE 2.14.0-3: build into an owned prefix

Clone the official repository and pin the latest published release. The library, VFIO/UIO plugins and utilities must come from the same installation.[\[17\]](#)

```
git clone https://github.com/OFS/opae-sdk.git "$OPAE_SRC"
git -C "$OPAE_SRC" checkout --detach \
    e4b329185d2c462f5b8c3c74443a5c5cdb25f3b3
git -C "$OPAE_SRC" rev-parse HEAD
# The supplied configuration-only patch makes private module selection relocatable.
git -C "$OPAE_SRC" apply --check \
    /path/to/manual-sources/opae214-relocatable-modules.patch
git -C "$OPAE_SRC" apply \
    /path/to/manual-sources/opae214-relocatable-modules.patch
```

```
export PIP_IGNORE_INSTALLED=1
cmake -S "$OPAE_SRC" -B "$OPAE_BUILD" \
  -DCMAKE_BUILD_TYPE=Release \
  -DCMAKE_INSTALL_PREFIX="$OPAE_PREFIX" \
  -DCMAKE_INSTALL_LIBDIR=lib64 \
  -DPython3_EXECUTABLE=/usr/bin/python3.9 \
  -DPYTHON_EXECUTABLE=/usr/bin/python3.9 \
  -DOPAE_BUILD_TESTS=OFF -DOPAE_BUILD_LEGACY=OFF \
  -DOPAE_BUILD_EXTRA_TOOLS=ON -DOPAE_BUILD_OPAEIO=ON \
  -DOPAE_BUILD_USERCLK=ON -DOPAE_BUILD_PLUGIN_VFIO=ON \
  -DOPAE_BUILD_LIBOPAEVFIO=ON -DOPAE_BUILD_LIBOPAEUIO=ON
cmake --build "$OPAE_BUILD" --parallel "$(nproc)"
cmake --install "$OPAE_BUILD"
```

The Python executable, development headers and libpython must use the same ABI. The supplied OP AE configuration patch makes private module lookup relocatable; it changes search paths, not programming functions. Activate the matching private runtime when using this build. Set `PIP_IGNORE_INSTALLED` to keep staged wheel builds from trying to uninstall host packages. If CMake selects an interpreter without its development package, set the matching `Python3_EXECUTABLE` and development paths or install that interpreter's development package. A fresh installation uses normal matching packages. An existing host can instead use signed, matching development RPMs extracted into an owned sysroot.

Confirm the prefix contains `fpgaconf`, `opae.io`, `pci_device`, `libopae-c`, its VFIO/UIO plugins, platform utilities and Python support. Do not run a hardware-enumerating tool as a substitute for inspecting installed files or ELF dependencies. The packaged ASP supplies a coherent private runtime for the verified application route; Section 5 selects it.

3.6 Linux DFL driver on Rocky 9.8

Use the reviewed `linux-dfl-backport` source pin `1d01b806f5bd7c65e2e46c37d43ba4aa6a4a047c`. The source requires the Rocky/RHEL 9.8 API compatibility patch supplied with this manual. Its changes cover the DFL bus-match signature, timer helper and two Silicom callback APIs; they do not enable new IA-840F networking features.^[6]

```
DFL_SRC="$WORK/src/linux-dfl-backport"
git clone https://github.com/OFS/linux-dfl-backport.git "$DFL_SRC"
git -C "$DFL_SRC" checkout --detach \
  1d01b806f5bd7c65e2e46c37d43ba4aa6a4a047c
rpm -q "kernel-devel-$(uname -r)"
readlink -f "/lib/modules/$(uname -r)/build"
git -C "$DFL_SRC" apply --check \
  /path/to/manual-sources/dfl-rocky98-compat.patch
git -C "$DFL_SRC" apply \
  /path/to/manual-sources/dfl-rocky98-compat.patch
make -C "$DFL_SRC" CONFIG_IOMMU_SVA=n
sudo make -C "$DFL_SRC" CONFIG_IOMMU_SVA=n install
sudo depmod -a
sudo modprobe dfl-pci
```

`CONFIG_IOMMU_SVA=n` omits the optional PASID/SVA helper not needed by this VFIO route. Before rebuilding an existing installation, check it: enterprise **weak-updates** may correctly reuse a kABI-compatible module even when its vermagic names the earlier kernel against which it was compiled.^[14]

```
modinfo -n dfl-pci
```

```
modinfo -F srcversion dfl-pci  
cat /sys/module/dfl_pci/srcversion  
lsmod | grep -E '^dfl|^fpga'
```

Installed/loaded identity, successful binding and functional tests take precedence over a guessed module path. Do not rebuild solely because the module resolves through **weak-updates**.

4 The OFS 2026.1 FIM and matching PIM platform

4.1 Use the paired IA-840F platform

The release consumes a previously implemented static FIM and a PIM template configured for it. These pins identify the donor tuple, not an unmodified upstream devkit project that automatically becomes an IA-840F.[7]

Component	Pin
OFS FIM, ofs-2026.1-1	866c25bb166810f65aae4f6b15374d0a89810e69
FIM common	147cae890b7d1245301cf5cde229f761b287b70d
PIM	3c21189e728009d4c492fa2be54c0ab1008b06dc
ASP donor	1af2ca74c452cb6ebbf54beb86e758e53489e826

The supplied `pr_build_template` contains `bin/` and `hw/lib/`, including `fme-ifc-id.txt`, `fme-platform-class.txt`, `platform/interface` databases and the preserved static netlist. PIM is already configured for `ofs_agilex`. [5]

```
cat "$PR/hw/lib/fme-ifc-id.txt"
cat "$PR/hw/lib/fme-platform-class.txt"
STATIC="$PR/hw/lib/build/syn/board/ia840f/syn_top/ofs_top.qdb"
sha256sum "$STATIC"
```

Require the FME UUID from Section 1 and static-QDB SHA-256 `dbc1684ab873b3d317d20019430c99177653daf221e7471b227b1966d7ef30b4`. [20] The matching base SOF, static MSF and PR PMSF also belong to this exact export; assembly cannot be repaired by borrowing masks from a different FIM.

4.2 Already-provisioned versus factory-programmed card

If the intended board already has the matching 2026.1 static FIM, verify its identity and skip permanent programming. Routine vector-add uses its `.gbs` PR image, not a full-device RPD. A factory card instead needs the separate, accepted full-device provisioning input and a safe activation plan. The ASP `.gbs`, an arbitrary SOF and a JIC are not interchangeable SDK writer inputs. [8]

The retained current-platform provisioning bundle has these accepted file identities. Obtain the files and their recovery material from the project release custodian; a public source checkout does not contain licensed images.

Artifact	Accepted SHA-256
Full-device SOF, 10,067,421 bytes	00dbd01b5b8c7c0b49fb1f9340ac9464e61a634ff045a1c0c4b79a464175d46f
SDK RPD, 10,670,080 bytes	96fb0dc9e0716a712f9e77a09b11161ac495a6e16436beb756bf7fd04874647e
MAP	818df1df710349adf3b6720508ea3a91481a25162bb3ec304fb4eba3dcc0ffc2

These describe a complete non-RSU `BOOT_INFO/P1` layout at address `0x00000000`: `BOOT_INFO` ends at `0x001FFFFF`; `P1` is `0x00200000..0x00A2CFFF`. They do not describe an RSU user-slot update at `0x04000000`. Do not silently change an existing board's flash scheme. [8]

4.3 Permanent provisioning: conditional operator procedure

Use `bw_agilex_flash_programmer`, not JTAG, for this project's persistent flash route. Bind the physical hardware revision, management PF/card index, USB BMC identity, current ownership, exact input hash, layout/erase footprint and recovery access before writing. Management PF1 and application PF0/VF0 serve different roles; do not create an application VF merely to flash.^[9]

For the reviewed complete-image input, after those prerequisites:

```
SDK_BIN="$HOME/.local/bin"    # Verify this SDK's actual installed path.
CARD_INDEX=<verified PCI management card index>
RPD="/absolute/path/to/migrated-caps03-sdk.rpd"
sha256sum "$RPD"
# Require the accepted RPD hash printed above before proceeding.
sudo systemctl start bittware-vfiod
"$SDK_BIN/bw_card_list" -i PCI
"$SDK_BIN/bw_agilex_flash_programmer" -i PCI -c "$CARD_INDEX" \
  program --force -a 0x00000000 "$RPD"
```

The original operation must return zero and complete erase, program, readback and built-in byte comparison. `--force` permits the reviewed slow VFIO transport; it is not a verification bypass. Do not run a concurrent BMC/status query, launch another writer, or activate while the writer is incomplete. The file's `bitswap=OFF` convention is paired with the reviewed writer's byte transformation; do not pre-reverse it.^[8]

4.4 Activation and postboot verification

A successful flash write leaves the running fabric intact. Activation requires supported quiescence and board/host-specific preparation, followed by separate BMC Off/On readbacks and one normal workstation reboot. The accepted deployment used a precisely bound card-only PCIe preparation; its historical root-port addresses/masks are not generic commands for another host. Broad AER disabling, blind removal/rescan and abrupt live power cuts are not this procedure.^[9]

Once the supported preparation and empty ownership are established, the BMC setters/readbacks use the verified USB device, not a guessed PCI card ordinal:

```
BMC_DEVICE=<USB device matched to this physical card>
"$SDK_BIN/bw_bmc_configure" -i USB -d "$BMC_DEVICE" power -p Off
"$SDK_BIN/bw_bmc_configure" -i USB -d "$BMC_DEVICE" power
# Require explicit OFF and native zero.
sleep 12
"$SDK_BIN/bw_bmc_configure" -i USB -d "$BMC_DEVICE" power -p On
"$SDK_BIN/bw_bmc_configure" -i USB -d "$BMC_DEVICE" power
# Require explicit ON and native zero.
sleep 30
sudo systemctl reboot
```

Record the preboot ID and completed receipts before reboot. Afterwards require a new boot ID, the intended PF0/management identities/bindings, restored PCIe policy and the migrated cached static FME identity. A missing application VF before per-boot preparation is expected. Deployment establishes the static platform; it does not prove vector-add arithmetic.^[9]

4.5 Rebuilding the FIM/PIM is a maintainer operation

For ordinary application development use the matching supplied netlist/PIM export. Rebuilding static FIM requires the exact IA-840F board overlay, current BittWare management integration, accepted memory/pin/clock/floorplan setup, licensed IP and prepared native CMake workspace. Cloning the FIM donor tag alone does not supply those inputs. Never substitute N6001 selectors or claim that a bare upstream checkout is a zero-step IA-840F implementation.[\[7\]](#) A new static implementation produces a new compatibility contract and needs its own platform export and hardware qualification before using these AFU images. Avoid rebuilding a qualified FIM solely to run vector-add.

5 Installing and selecting the IA-840F ASP

5.1 Extract and verify the release

Select the archive whose release metadata names OPAE 2.14.0-3 and the qualified FIM/Quartus/AHLS tuple. Verify the supplied archive checksum before extracting into a fresh user-owned directory:

```
ARCHIVE="/absolute/path/to/the/current-ASP-release.tar.gz"
sha256sum "$ARCHIVE"
mkdir -p "$WORK/asp"
tar -xzf "$ARCHIVE" -C "$WORK/asp"
test -f "$BSP/board_env.xml"
test -f "$BSP/manifest.json"
```

Do not merge into another installed BSP prefix. Manifest verification must check files, sizes, hashes and internal links, rather than merely counting files or trusting extraction exit zero. The published release metadata and checksum accompany this manual.

5.2 Directory map

Directory/file	Role
board_env.xml, hardware/	Compiler environment, two variants, device model and shared RTL.
board_local/	Reviewed board-specific overrides; shared donor originals remain available.
pr_build_template/	Matching static/PIM/compiler export.
cmake/	Automatic native BSP backend and build support.
source/, software/	MMD/MPF source and board-local software changes.
linux64/ and private OPAE prefix	Coherent runtime, utilities and plugins.
bringup/binaries/	Four named SYCL .fpga executables.
bringup/images/, bringup/aocxs/	Matching GBS programming images and compiler containers.
bringup/prebuilt/	Native import archives containing the original program and fitted image.
examples/vector-add/	Migrated CMake project and original sample sources.
qualification/, RELEASE.json	Runtime/image qualification, source pins and limits.

The default <variant>.fpga aliases select afu_flat; kclk products have distinct names. Do not overwrite one flow's image with another.

5.3 Compiler environment: source AHLS, then select Quartus

```
source "$AHLS/fpgavars.sh" >/dev/null
export INTELFPGAOCSDKROOT="$AHLS"
export ALTERA_HLS_ROOT="$AHLS"
export QUARTUS_ROOTDIR="$QUARTUS"
export QUARTUS_ROOTDIR_OVERRIDE="$QUARTUS"
export PATH="$QUARTUS/bin:$QUARTUS/sopc_builder/bin:$AHLS/bin:$PATH"
export OPAE_PLATFORM_ROOT="$PR"
export OPAE_PLATFORM_FPGA_FAMILY=AGILEX
export QUARTUS_VERSION=26.1 QUARTUS_VERSION_MAJOR=26
```

```
export PR_COMPILE=1 BUILD_ROOT_REL=../../../../..
ulimit -v 57671680
command -v ahls quartus_sh
ahls --version
quartus_sh --version
```

Setting Quartus after `fpgavars.sh` avoids a mixed tool generation. `QUARTUS_ROOTDIR_OVERRIDE` can redirect even a fully spelled executable path, so verify its actual version rather than only its pathname.

5.4 Private runtime activation

The release's `activate.sh` selects its own library/plugin paths and writes prefix-specific FCD, ICD and OPAE configuration files within that prefix. Source it after the compiler environment:

```
source "$BSP/activate.sh"
printf '%s\n' "$LD_LIBRARY_PATH" "$ACL_BOARD_VENDOR_PATH" \
"$OCL_ICD_VENDORS" "$LIBOPAE_CFGFILE"
```

This is not a global `aoclinstall`; no system-wide FCD/ICD registration is needed. Activation is a setup operation, not a DMA test or PR load. Use the release's matching OPAE tools and plugins together. An ELF loader check can verify library resolution without executing an application main, but the final numerical run must also confirm the actual loaded runtime.

6 Preparing the application VF and card access

6.1 Discover, do not copy, hardware identifiers

Identify the physical IA-840F and its actual FME, management and VF functions. The qualification example used FME 0000:4f:00.0, management PF 0000:4f:00.1 and application VF 0000:4f:00.2. Bus numbers, IOMMU groups and card indices can change. PCI ID text may say N6001 because of inherited IDs; it is not sufficient physical board identification.^[10]

```
lspci -Dnn
FME_BDF="<verified FME BDF>"
PF_SYS="$(readlink -f "/sys/bus/pci/devices/$FME_BDF")"
readlink -f "$PF_SYS/driver"
cat "$PF_SYS/vendor" "$PF_SYS/device" "$PF_SYS/subsystem_device"
cat "$PF_SYS/sriov_numvfs" "$PF_SYS/sriov_drivers_autoprobe"
```

Require PF0 bound to `df1-pci`, the intended static FME interface and no competing device holders/maps before creating a VF or programming an AFU. On the qualified driver the child PR region's `compat_id` is a cached probe-time identifier; do not assume every driver's similarly named accessor is an ordinary-file read.

6.2 Per-boot VF creation and explicit VFIO selection

After reboot there may be no VF and `sriov_numvfs=0`. That is a setup state, not a failed FIM boot. Disable *per-PF* automatic probing before creating the intended VF; otherwise a matching driver may immediately access its AFU before clock/reset readiness is established.^[10]

```
# Only when no application/VF/programmer owner exists and no VF is present:
AUTOPROBE_BEFORE="$(cat "$PF_SYS/sriov_drivers_autoprobe")"
printf 0 | sudo tee "$PF_SYS/sriov_drivers_autoprobe" >/dev/null
pci_device "$FME_BDF" vf 1 # Use the selected OPAE tool; privileged access may be needed.
VF_SYS="$(readlink -f "$PF_SYS/virtfn0")"
VF_BDF="$(basename "$VF_SYS")"
readlink -f "$VF_SYS/physfn"
cat "$VF_SYS/vendor" "$VF_SYS/device"
readlink -f "$VF_SYS/iommu_group"
# Require the intended VF is unbound and physfn resolves to PF_SYS.
printf vfio-pci | sudo tee "$VF_SYS/driver_override" >/dev/null
sudo opae.io init -d "$VF_BDF" "$(id -un):$(id -gn)"
readlink -f "$VF_SYS/driver"
# Require vfio-pci, correct group membership, and unchanged management PF.
printf '%s' "$AUTOPROBE_BEFORE" | \
    sudo tee "$PF_SYS/sriov_drivers_autoprobe" >/dev/null
```

Run the privileged OPAE command with the selected prefix's environment, as in Section 8; plain `su` do may discard library/Python selectors. A failed or unknown command stops the sequence. Restore autoprobe only after successful, verified binding; do not blindly remove or re-create an existing VF. The qualified application VF is a singleton IOMMU group. Group numbers are not a universal constant.^[10]

6.3 Memory lock, huge pages and ownership

Check memory-lock limits in the environment that will actually run the program:

```
ulimit -l  
grep -E '^HugePages_|^Hugepagesize:' /proc/meminfo
```

The initial qualified route uses a supported privileged runtime operation, not global permission changes made merely to obtain a pass. For a non-root production configuration, deliberately provision the required permissions, memlock/huge-page policy and group access, then validate that configuration.

Ownership checks need the actual group/device paths and *root-visible* file descriptors and memory maps. A process-name search or absence of a GUI window is not proof of empty ownership. Preserve an uncertain application's queue, event, allocations, pins and descriptors; do not force teardown while posted writes or DMA retirement are unresolved.

7 Building vector-add through the BSP

7.1 Select one coherent build configuration

The standard variants use `vector-add-buffers.cpp`; USM variants use `vector-add-usm.cpp`. The project preserves the original sample and adapts only the older Intel FPGA header-selector dialect in a build-local copy. The addition lambdas and full CPU oracles are retained.^[11]

Start with:

```
VARIANT=ofs_ia840f
FLOW=afu_flat
BUILD="$WORK/build/vector-add-${VARIANT}-${FLOW}"
```

Use a different build directory for each variant/flow. The `kclk` macro changes integration topology; do not reuse a mutable prepared project across flows.

7.2 Normal full hardware compilation

```
cmake -S "$BSP/examples/vector-add" -B "$BUILD" \
  -DAHLS_ROOT="$AHLS" -DQUARTUS_ROOT="$QUARTUS" \
  -DASP_ROOT="$BSP" -DASP_VARIANT="$VARIANT" -DASP_FLOW="$FLOW" \
  -DOPAE_PLATFORM_ROOT="$PR" -DASP_RUNTIME_ROOT="$BSP" \
  -DAHLS_MEMORY_LIMIT_GIB=55
cmake --build "$BUILD" --target report --verbose
# Inspect report/RTL estimates; this is not a fitted FPGA executable.
cmake --build "$BUILD" --target fpga --verbose
```

The report uses `-fsycl-link=early`. The full `fpga` target uses the actual AHLS hardware linker and the selected BSP. It can take hours and must complete integration generation, synthesis, fit, final timing, assembly, resource reporting and image/container creation. A front-end exit or generated report alone is not completion.^[11]

7.3 The native compile/link pattern

For a source already using the AHLS dialect:

```
ahls -DFPGA_HARDWARE -c vector-add-buffers.cpp \
  -o vector-add-buffers.o
ahls -Xshardware -Xstarget="$BSP:ofs_ia840f" \
  -Xsbsp-flow=afu_flat vector-add-buffers.o \
  -o vector-add-buffers.fpga
```

The BSP target belongs at hardware link time. Do not substitute a part-only target for a board build. Let the BSP callbacks handle native Quartus work, clock optimization/selection, final STA and image creation. Do not derive clocks in separate vector-add scripts or edit a frequency by hand to conceal a timing failure.^[11]

The port retains the vendor two-stage clock policy: an optimization request is distinct from the operating clock admitted by final timing and GBS metadata. The distributed image frequencies are not guaranteed frequencies for arbitrary new kernels. Changing device code requires its own compile and numerical

qualification.

7.4 Verified prebuilt-image link: fastest initial bring-up

Use this route when the example is unchanged. It imports the original host program, symbol/property metadata and validated AOCX through the native AHLS linker; it does not recompile changed device code or start a new FPGA fit.

```
BUILD="$WORK/build/vector-add-prebuilt-${VARIANT}-${FLOW}"
PREBUILT="$BSP/bringup/prebuilt/${VARIANT}__${FLOW}.fpga.a"
cmake -S "$BSP/examples/vector-add" -B "$BUILD" \
  -DAHLS_ROOT="$AHLS" -DQUARTUS_ROOT="$QUARTUS" \
  -DASP_ROOT="$BSP" -DASP_VARIANT="$VARIANT" -DASP_FLOW="$FLOW" \
  -DOPAE_PLATFORM_ROOT="$PR" -DASP_RUNTIME_ROOT="$BSP" \
  -DAHLS_PREBUILT_ARCHIVE="$PREBUILT"
cmake --build "$BUILD" --target fpga --verbose
```

The verified native imports reproduce each distributed hardware-qualified executable byte-for-byte. The opaque import archive already contains its host program; linking another `main` or silently ignoring changed source would be an error. Use only the archive matching the selected variant/flow. The prebuilt mode is exclusive with other reuse options.^[11]

7.5 Find the result and inspect it

Standard output names are:

```
vector-add-buffers__ofs_ia840f__afu_flat.fpga
vector-add-buffers__ofs_ia840f__afu_flat_kclk.fpga
vector-add-usm__ofs_ia840f__usm__afu_flat.fpga
vector-add-usm__ofs_ia840f__usm__afu_flat_kclk.fpga
```

The executable is a SYCL fat binary containing its matching FPGA image and compiler metadata. A bare AOCX, raw GBS or early-link `.fpga.a` report is not an interchangeable executable. AHLS may move its working project into the final `.fpga.prj`; do not report a missing old temporary path as failed hardware compilation.

8 Loading and running the FPGA program

8.1 Pre-run checklist

Before either an explicit PR load or a SYCL program that can program the device, require the matching static FIM, exact intended card/VF bindings, correct variant/flow/image, healthy setup and empty root-visible ownership. Store the current boot ID, image/executable hashes and the operation's original native exit. Do not run two application owners or programming operations concurrently.

For the packaged standard/plain image, after identifying the intended FME:

```
VARIANT=ofs_ia840f
FLOW=afu_flat
GBS="$BSP/bringup/images/${VARIANT}__${FLOW}.gbs"
EXE="$BSP/bringup/binaries/${VARIANT}__${FLOW}.fpga"
sha256sum "$GBS" "$EXE"
```

For a full fresh build, use that build's *own* emitted GBS and executable. Do not load a packaged image and claim the freshly compiled, different image was verified.

8.2 Preserve the selected runtime under privilege

Use a clean privileged shell to source the explicit AHLS/BSP environments rather than relying on `sudo -E` to preserve every unrelated user variable. The following supported PR pattern does not bypass clock configuration:

```
sudo bash --noprofile --norc -c '
  source "$1/fpgavars.sh" >/dev/null
  source "$2/activate.sh"
  exec fpgaconf "$3" "$4"
' ia840f "$AHLS" "$BSP" "$FME_BDF" "$GBS"
```

`fpgaconf` resolves through the selected OPAE prefix from activation. Require native zero and the intended resulting AFU identity before the application. A command that explicitly names an unrelated `/usr/bin` utility may select the wrong installation even when its libraries appear in `LD_LIBRARY_PATH`.

Then run the actual fat binary:

```
sudo bash --noprofile --norc -c '
  source "$1/fpgavars.sh" >/dev/null
  source "$2/activate.sh"
  unset CL_CONTEXT_MPSIM_DEVICE_INTELFPGA
  unset CL_CONTEXT_COMPILER_MODE_INTELFPGA
  unset CL_CONTEXT_EMULATOR_DEVICE_ALTERA
  exec "$3" 10000
' ia840f "$AHLS" "$BSP" "$EXE"
```

This selects real FPGA hardware through the source-defined Altera hardware selector. Clear stale emulator/simulator settings; a successful emulator run is not hardware validation.

8.3 CMake run target

For a configured build with the matching image already loaded:

```
cmake -S "$BSP/examples/vector-add" -B "$BUILD" \
  -DVECTOR_ADD_RUN_WITH_SUDO=ON -DVECTOR_ADD_SIZE=10000
cmake --build "$BUILD" --target run_fpga --verbose
cat "$BUILD/native-runtime-exit.txt"
```

The run target records the original application exit separately. Existing configuration/path/reuse options remain in the build cache; use a dedicated build directory and inspect it before reconfiguration. `run_fpga` depends on `fpga`, so an out-of-date image can trigger a rebuild.

8.4 What success means

The source computes the sequential CPU sum and compares every output element before reporting success. The printed first/last elements are display examples, not the whole oracle. The required conditions are hardware device selection, full array agreement, native exit zero, correct image/runtime identity and accepted post-run ownership/kernel diagnostics.[\[11\]](#)

The numerical output has this form; the physical device-name suffix is installation-dependent:

```
Running on device: ofs_ia840f : Intel OFS Platform (...)
Vector size: 10000
[0]: 0 + 0 = 0
[1]: 1 + 1 = 2
[2]: 2 + 2 = 4
...
[9999]: 9999 + 9999 = 19998
Vector add successfully completed on device.
```

The original sample assumes at least three elements when printing its display indices. Do not use size zero, one or two as a supported example invocation.

9 Validating all four configurations

9.1 Use distinct products and one owner at a time

Variant	Flow	Source/memory path
ofs_ia840f	afu_flat	Buffer/local DDR vector-add
ofs_ia840f	afu_flat_kclk	Same buffer kernel, alternate integration clock topology
ofs_ia840f_usm	afu_flat	Shared-USM vector-add
ofs_ia840f_usm	afu_flat_kclk	Same USM kernel, alternate integration clock topology

Configure each row separately and select its own `.gbs/.fpga` pair. A pass for one variant or clock flow does not qualify another. Reuse unchanged, verified images instead of refitting merely to repeat numerical validation.

9.2 Qualification record for this release

OPAE 2.14.0-3 hardware regression: all four rows passed.

The final project runtime used the released source pin and the documented module-search configuration patch. Each row ran the unchanged SYCL fat binary and its own matching GBS, with rebuilt MMD/MPF and private OPAE core/plugins.[\[19\]](#)

Variant	Flow	MHz	Comparisons	Native exit
ofs_ia840f	afu_flat	618	10000	0
ofs_ia840f	afu_flat_kclk	490	10000	0
ofs_ia840f_usm	afu_flat	592	10000	0
ofs_ia840f_usm	afu_flat_kclk	479	10000	0

All 40,000 comparisons passed. The loaded SYCL/OpenCL/FPGA-runtime bytes matched AHLS; the OPAE core and both selected plugins resolved inside the private release prefix. Every application ended with no device holders or unreadable owner entries. No FPGA synthesis, fitting, flash or reboot was needed for this software upgrade.[\[19\]](#)

The displayed MHz values are operating-clock STA/GBS metadata, not a live frequency or throughput measurement. The exact scoped VF pending-before-FLR warning remains recorded. The verdict is `lifecycle_clean=false`; no universal reset/write-retirement claim is added.[\[19\]](#)

9.3 Optional short numerical probes

The example's runtime size argument can exercise odd-length transfers. Use fresh, bounded processes, the same qualified executable/image and the same ownership/runtime rules:

```
# Under the selected privileged AHLS/BSP runtime environment:
"$EXE" 3
"$EXE" 17
"$EXE" 257
"$EXE" 10003
# Buffer example only: three queued submissions, one full final oracle.
"$EXE" 10003 3
```

Each run still checks every output. For the buffer example, the second argument changes the number of submissions; it is not independent validation of every intermediate output. Probe correctness is not a throughput benchmark. Do not run more probes after a failed/unknown hardware state until it is reconciled.

10 Debugging and known operating limits

10.1 Wrong Quartus generation or missing FPGA header

Inspect `QUARTUS_ROOTDIR_OVERRIDE`, actual tool resolution and the version banner after sourcing AHLS. The qualified header is `sycl/ext/altera/fpga_extensions.hpp`; older sample paths use the Intel namespace. Search the installed tree if the expected header is missing rather than mixing headers from another compiler generation.

10.2 Python development mismatch during OPAE configuration

A found interpreter is not enough to build `opae.io`/Python bindings. Require the matching interpreter, `Python.h` and shared `libpython`. Explicit `Python3_EXECUTABLE`/development paths prevent CMake from selecting an installed interpreter whose development package is absent. Missing libedit headers similarly need their correct development package, not a suppressed compile error.

10.3 Cannot open the device or user-clock UIO node

First check the intended VF binding, IOMMU group, memory-lock/huge-page policy and environment. A non-root user-clock access failure is a setup failure, not proof the FPGA addition is wrong. Use the documented privileged path while qualifying setup instead of changing global device permissions. Clear simulator context variables before physical FPGA work.

10.4 No VF after reboot

Verify the matching FIM/PF0 first, then perform Section 6. Absence of `virtfn0` before per-boot creation is expected. Do not interpret it as a reason for another flash, power cycle or reboot.

10.5 Timing failure, ROM failure or missing base outputs

Final operating-clock timing must pass; do not hide negative slack by hand-editing fixed static clock requirements. A source-generated `sys_description.hex` initializer belongs to the actual kernel and must be present before synthesis. An all-zero runtime configuration ROM can occur even when tools return zero if a missing initializer was zero-filled. Reject missing/zero-fill warnings. The matching base SOF/MSF/PMSF are required for PR assembly; a recovered base mask does not authorize borrowing another platform's collateral.

10.6 Native reuse declines the image

A raw AOCX is not a native `-reuse-exe` executable anchor, and an executable anchor can still be rejected by the toolchain's identity checks. For this unchanged example use the verified native `AHLS_PREBUILT_ARCHIVE` mode. Do not let a supposedly host-only rebuild silently start a new FPGA fit. An imported archive preserves its original program; it is not a way to claim changed device code was compiled.

10.7 The retained VF lifecycle warning

The exact scoped warning previously accepted for this card is:

```
vfio-pci 0000:4f:00.2: timed out waiting for pending transaction;  
performing function level reset anyway
```

The hardware/native result, empty ownership and kernel warning are recorded separately. This disposition does not erase the warning or imply universally clean write retirement/reset handling. Do not generalize it to another BDF, AER/IOMMU fault, failed numerical result or uncertain owner. This release's actual OPAE 2.14.0-3 observations are in [Section 9](#).

10.8 Failure and recovery discipline

On an error or deadline, stop further device actions. Preserve the original native exit/log and the live owner's PID/start/boot identity. Do not kill an uncertain owner, free pins, PR another image or launch another probe merely because a watcher timed out. A specifically approved recovery uses the established supported board/host sequence and is recorded as recovery, not as a passed numerical test. AER suppression, BAR scans, guessed UUID/register offsets and blind PCI rescans are not troubleshooting shortcuts.

11 Maintenance, package integrity and source boundaries

11.1 Keep a coherent runtime

Pin OPAE 2.14.0-3, build its library/plugins/utilities together, and build MMD/MPF against those headers/libraries. Select that same cohort at runtime. A system utility and a privately copied library with a familiar SONAME do not establish a coherent installation. Keep the current SDK SYCL/OpenCL libraries bound separately from the OPAE version.[\[17\]](#)

11.2 Rebuilding MMD/MPF

The BSP provides an explicit software target. It builds current-source MPF, MMD and utilities into an owned prefix, without a hardware query or global registration. The primary inputs are the actual AHLS SDK, accepted PR platform, OPAE installation and MPF source.[\[11\]](#)

```
SW_BUILD="$WORK/build/asp-software"
cmake -S "$BSP" -B "$SW_BUILD" \
  -DIA840F_PACKAGE_ROOT="$BSP" \
  -DIA840F_AHLS_ROOT="$AHLS" -DIA840F_QUARTUS_ROOT="$QUARTUS" \
  -DIA840F_PLATFORM_ROOT="$PR" -DIA840F_OPAE_ROOT="$OPAE_PREFIX" \
  -DIA840F_BBB_ROOT="$BSP/source/extra/intel-fpga-bbb"
cmake --build "$SW_BUILD" --target software --verbose
```

Build MPF before the dependent MMD targets. Verify cache-resolved headers and libraries, ELF dependencies and the actual loaded paths; copying an untested library over the installed prefix is not this process.

11.3 Image naming and archive validation

Keep four independently named flow products and plain default aliases. Preserve the matching executable, GBS, AOCX, resource metadata and source manifest as one set. Verify the archive's file contents and relative-link closure after packaging; archive exit zero alone is not content acceptance. Rebase only documented private FCD/ICD/configuration files after relocation.

11.4 What is outside the validated matrix

A different kernel, host OS, card topology, static FIM, FPGA tool version or allocation/traffic pattern needs its own applicable checks. The four vector-add rows do not establish all upstream samples, host pipes, HSSI/network variants, MTBF, universal cold/reset recovery or cross-distribution compatibility. Operating clock metadata is not a measured latency/throughput result.

12 Operator checklist

1. Obtain the current licensed tools, latest-release OPAE source, ASP archive, matched FIM/PR provisioning inputs and board-revision documentation.
2. Install/power/cool the card according to its hardware guide; configure IOMMU and SR-IOV deliberately.
3. Install matching host development packages and the qualified DFL driver; validate installed/loaded identity and PF0 binding.
4. Install Quartus 26.1.1/Agilex 7 and AHLS 2026.1.0; verify actual versions.
5. Build/select coherent OPAE 2.14.0-3 library/plugins/tools and matching MMD/MPF.
6. Verify the current static FIM. Provision only if necessary, using the complete accepted image and separately supported activation/recovery plan.
7. Extract and hash-check the BSP into a fresh prefix; source its private runtime.
8. Discover the actual FME/management/VF identities, prove empty ownership, and perform per-boot VF preparation when necessary.
9. Configure one variant/flow in a unique build directory. Use native hardware `-Xstarget` linking, or import its unchanged verified prebuilt program.
10. Require completed timing/image/resource metadata, load the exact matching GBS through the supported route, and run the actual SYCL fat binary.
11. Require the full numerical oracle and native exit zero; inspect ownership and the kernel journal with only the precise retained limitation allowed.
12. Repeat for each requested row, preserving individual result/image/runtime identities. Stop on failure/unknown state; do not escalate automatically.

Sources and qualification evidence

Public documents describe tool/architecture contracts. Local records establish this board port and actual qualification; they are not vendor certification. The official AHLS installation page was available through its indexed official text when the live documentation application did not render reliably. The source bundle retains the source ledger and the driver patch.

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